DH Matrix

$$\begin{pmatrix} \cos\theta_i & -\cos\alpha_i\sin\theta_i & \sin\alpha_i\sin\theta_i & a_i\cos\theta_i \\ \sin\theta_i & \cos\alpha_i\cos\theta_i & -\sin\alpha_i\cos\theta_i & a_i\sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- a_i is the distance DO_i oriented as x_i, is always constant and is the length of the common normal that intersects the joint axis i and $i+1$. We recall that the point D is the point of the intersection between the common normal and the joint's axis $i-1$.
- d_i is the distance $O_{i-1}D$, oriented as z_{i-1}, this value is variable if the joint i is prismatic
- α_i is the twist angle from z_{i-1} to z_i around x_i, is constant.
- θ_i is the angle from x_{i-1} to x_i around z_{i-1}, this value is variable if the joint i is revolute.

Jacobian Pseudo Inverse

pseudo inverse if n>m

$$J^{\#} = J^T(JJ^T)^{-1}$$

satisfies

$$JJ^{\#}J = J$$
$$J^{\#}JJ^{\#} = J^{\#}$$
$$(JJ^{\#})^T = JJ^{\#}$$
$$(J^{\#}J)^T = J^{\#}J$$

pseudo inverse if m>n

$$J^{\#} = (J^TJ)^{-1}J^T$$

if J^TJ or JJ^T are singular the matrix is computed with SVD

Weigthed Pseudo Inverse

Given the problem $J\dot{\mathbf{q}} = \dot{\mathbf{y}}$, and a positive definite matrix W, the solution

$$\dot{\mathbf{q}}_W = J_W^{\#}\dot{\mathbf{y}}$$

where

$$J_W^{\#} = W^{-1}J^T(JW^{-1}J^T)^{-1}$$

minimizes the weighted norm

$$\|\dot{\mathbf{q}}\|_W^2 = \dot{\mathbf{q}}^TW\dot{\mathbf{q}}.$$

$J_W^{\#}$ satisfies the relations of the pseudo inverse, except for $(J^{\#}J)^T = J^{\#}J$.

Skew Symm. Matrix

the skew symm. matrix of a vector:

$$\mathbf{v} = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \longrightarrow S = \begin{pmatrix} 0 & -v_z & v_y \\ v_z & 0 & -v_x \\ -v_y & v_x & 0 \end{pmatrix}$$

$$\mathbf{v} \times \mathbf{u} = S(\mathbf{v})\mathbf{u}$$

Linear Quadratic Optimization

$$\min_{\mathbf{x}\in\mathbb{R}^n} H(\mathbf{x}) = \frac{1}{2}(\mathbf{x}-\mathbf{x}_N)^TW(\mathbf{x}-\mathbf{x}_N)$$
$$\text{subject to} \quad J\mathbf{x} = \mathbf{b}$$

solution $\longrightarrow$ $\mathbf{x} = \mathbf{x}_N + W^{-1}J^T(JW^{-1}J^T)^{-1}(\mathbf{b} - J\mathbf{x}_N)$

some times the problem is written in a different form

$$x^TWx + b^Tx = (x-c)^TW(x-c)$$

the relation between b and c is $\longrightarrow$

$$b = -2Wc$$
$$c = -\frac{1}{2}W^{-1}b$$

Projected Gradient

we solve by choosing as preferred direction q_N the gradient of a function to maximize (if we want to minimize, we choose the gradient times -1)

$$\dot{\mathbf{q}} = J_W^{\#}\dot{\mathbf{y}} + (I - J_W^{\#}J)\dot{\mathbf{q}}_N \qquad \dot{\mathbf{q}}_N = \nabla G$$

Saturation in Null Space

$$\dot{\mathbf{q}}_{SNS} = (JW)^{\#}s\dot{\mathbf{y}} + \left(I - (JW)^{\#}J\right)\dot{\mathbf{q}}_N$$

- W is a diagonal matrix such that the entries are 0 or 1 and is needed to select which joint will saturate. If we want to saturate the joint i, then $W_{ii} = 0$, else, if the joint is "free" $W_{ii} = 1$.
- s is the scaling factor of the required task velocity $\dot{\mathbf{y}}$.
- $\dot{\mathbf{q}}_N$ is the vector of the saturated (full speed) joint velocities.

Coriolis/Centrifugal vector

Definition 3.1.3 : Christoffel symbols

The **Christoffel symbols** of the first kind associated to the inertia matrix of the robot $M(\mathbf{q})$ are a set of coefficients indexed by $i, j, k \in \{1 \dots n\}$. The Christoffel symbol c_{kij} is

$$c_{kij} = \frac{1}{2}\left(\frac{\partial m_{kj}}{\partial q_i} + \frac{\partial m_{ki}}{\partial q_j} - \frac{\partial m_{ij}}{\partial q_k}\right) \tag{3.67}$$

The symbols are symmetric: $c_{kji} = c_{kij}$.

$$c_k(\mathbf{q}, \dot{\mathbf{q}}) = \dot{\mathbf{q}}^T C_k(\mathbf{q}) \dot{\mathbf{q}}$$

$$C_k(\mathbf{q}) = \frac{1}{2}\left(\frac{\partial M_k}{\partial \mathbf{q}} + \left(\frac{\partial M_k}{\partial \mathbf{q}}\right)^T - \frac{\partial M}{\partial q_k}\right)$$

La matrice di fattorizzazione che soddisfa le prorpietà di skew simmetria ha la i-esima riga definita come segue

$$S_i^T = \dot{q}^T C_i(q) \Rightarrow S = \begin{bmatrix} S_1^T \\ \vdots \\ S_n^T \end{bmatrix} \Rightarrow \dot{M} - 2S \quad \text{skew symmetric}$$

Algorithm for Kinetic Energy

$$\sigma_i = \begin{cases} 0 \text{ if joint } i \text{ is revolute} \\ 1 \text{ if joint } i \text{ is prismatic} \end{cases}$$

$${}^i\boldsymbol{\omega}_i = {}^{i-1}R_i^T(q_i)[{}^{i-1}\boldsymbol{\omega}_{i-1} + (1-\sigma_i)\dot{q}_i\,{}^{i-1}\mathbf{z}_{i-1}]$$

$${}^i\mathbf{v}_i = {}^{i-1}R_i^T(q_i)[{}^{i-1}\mathbf{v}_{i-1} + \sigma_i\dot{q}_i\,{}^{i-1}\mathbf{z}_{i-1} + {}^{i-1}\boldsymbol{\omega}_i \times {}^{i-1}\mathbf{r}_{i-1,i}]$$

$$\mathbf{v}_{ci} = \mathbf{v}_i + \boldsymbol{\omega}_i \times \mathbf{r}_{ci}$$

$\mathbf{r}_{ci}$: is the CoM in the reference RF_i and is constant

${}^{i-1}\mathbf{r}_{i-1,i}$: is the last column of the DH matrix from i-1 to i

Task Priority

- start with the first task and reformulate the problem so as to provide always a "solution", at least in terms of minimum error norm

for $i = 1$

$$\begin{cases} \dot{\boldsymbol{q}}_1 = \arg\min_{\dot{\boldsymbol{q}} \in \mathbb{R}^n} \frac{1}{2}\|\dot{\boldsymbol{q}}\|^2 \\ \text{s.t.} \quad \boldsymbol{J}_1\dot{\boldsymbol{q}} = \dot{\boldsymbol{y}}_1 \end{cases} \Rightarrow \begin{cases} \dot{\boldsymbol{q}}_1 = \arg\min_{\dot{\boldsymbol{q}} \in \mathcal{S}_1} \frac{1}{2}\|\dot{\boldsymbol{q}}\|^2 \\ \mathcal{S}_1 = \left\{\arg\min_{\dot{\boldsymbol{q}} \in \mathbb{R}^n} \frac{1}{2}\|\boldsymbol{J}_1\dot{\boldsymbol{q}} - \dot{\boldsymbol{y}}_1\|^2\right\} \end{cases}$$

$$\Rightarrow \dot{\boldsymbol{q}}_1 = \boldsymbol{J}_1^{\#}\dot{\boldsymbol{y}}_1 \Rightarrow \mathcal{S}_1 = \{\dot{\boldsymbol{q}}_1 + \boldsymbol{P}_1\boldsymbol{v}_1,\ \boldsymbol{v}_1 \in \mathbb{R}^n\}$$

for $i = 2$

$$\begin{cases} \dot{\boldsymbol{q}}_2 = \arg\min_{\dot{\boldsymbol{q}} \in \mathcal{S}_2} \frac{1}{2}\|\dot{\boldsymbol{q}}\|^2 \\ \mathcal{S}_2 = \left\{\arg\min_{\dot{\boldsymbol{q}} \in \mathcal{S}_1} \frac{1}{2}\|\boldsymbol{J}_2\dot{\boldsymbol{q}} - \dot{\boldsymbol{y}}_2\|^2\right\} \end{cases} \Rightarrow \begin{aligned} \dot{\boldsymbol{q}}_2 &= \dot{\boldsymbol{q}}_1 + (\boldsymbol{J}_2\boldsymbol{P}_1)^{\#}(\dot{\boldsymbol{y}}_2 - \boldsymbol{J}_2\dot{\boldsymbol{q}}_1) \\ \mathcal{S}_2 &= \{\dot{\boldsymbol{q}}_2 + \boldsymbol{P}_{A,2}\boldsymbol{v}_2,\ \boldsymbol{v}_2 \in \mathbb{R}^n\} \end{aligned}$$

for the generic step $i = k$

$\dot{\boldsymbol{q}}_{k-1}$ — prioritized solution up to task $k-1$

$\mathcal{S}_{k-1} = \{\dot{\boldsymbol{q}}_{k-1} + \boldsymbol{P}_{A,k-1}\boldsymbol{v}_{k-1},\ \boldsymbol{v}_{k-1} \in \mathbb{R}^n\}$ — set of all solutions up to task $k-1$

LQ problem for k-th task

$$\begin{cases} \dot{\boldsymbol{q}}_k = \arg\min_{\dot{\boldsymbol{q}} \in \mathcal{S}_k} \frac{1}{2}\|\dot{\boldsymbol{q}}\|^2 \\ \mathcal{S}_k = \left\{\arg\min_{\dot{\boldsymbol{q}} \in \mathcal{S}_{k-1}} \frac{1}{2}\|\boldsymbol{J}_k\dot{\boldsymbol{q}} - \dot{\boldsymbol{y}}_k\|^2\right\} \end{cases}$$

recursive formula (Siciliano, Slotine: ICAR 1991)

$$\boxed{\dot{\boldsymbol{q}}_k = \dot{\boldsymbol{q}}_{k-1} + (\boldsymbol{J}_k\boldsymbol{P}_{A,k-1})^{\#}(\dot{\boldsymbol{y}}_k - \boldsymbol{J}_k\dot{\boldsymbol{q}}_{k-1})}$$

prioritized solution up to task k

correction needed when the solution up to task $k-1$ is not satisfying also task k

initialization: $\dot{\boldsymbol{q}}_0 = \boldsymbol{0}$, $\boldsymbol{P}_{A,0} = \boldsymbol{I}$

over the steps, the search set is progressively reduced $\Leftrightarrow$ $\mathbb{R}^n = \mathcal{S}_0 \supseteq \mathcal{S}_1 \supseteq \cdots \supseteq \mathcal{S}_{s-1} \supseteq \mathcal{S}_s$

- the solution considering the first k tasks with their priority

$$\dot{\boldsymbol{q}}_k = \dot{\boldsymbol{q}}_{k-1} + (\boldsymbol{J}_k\boldsymbol{P}_{A,k-1})^{\#}(\dot{\boldsymbol{y}}_k - \boldsymbol{J}_k\dot{\boldsymbol{q}}_{k-1})$$

satisfies also ("does not perturb") the previous $k-1$ tasks

$$\boldsymbol{J}_{A,k-1}\dot{\boldsymbol{q}}_k = \boldsymbol{J}_{A,k-1}\dot{\boldsymbol{q}}_{k-1}$$

since

$$\boldsymbol{J}_{A,k-1}\underbrace{(\boldsymbol{J}_k\boldsymbol{P}_{A,k-1})^{\#}} = \boldsymbol{J}_{A,k-1}\underbrace{\boldsymbol{P}_{A,k-1}(\boldsymbol{J}_k\boldsymbol{P}_{A,k-1})^{\#}} = \boldsymbol{O}$$

(Maciejewski, Klein: IJRR 1985): check the four defining properties of a pseudoinverse

- recursive expression also for the null-space projector

$$\boxed{\boldsymbol{P}_{A,k} = \boldsymbol{P}_{A,k-1} - (\boldsymbol{J}_k\boldsymbol{P}_{A,k-1})^{\#}\boldsymbol{J}_k\boldsymbol{P}_{A,k-1}} \qquad \boldsymbol{P}_{A,0} = \boldsymbol{I}$$

(Baerlocher, Boulic: IROS 1998): for the proof, see Appendix A

- when the k-th task is (close to be) incompatible with the previous ones (algorithmic singularity), use "DLS" instead of "#" in k-th solution...